

Learning Criterion

C6. Calculates Expected Monetary Value using stated or estimated probabilities and applies the Bayes decision criterion to select the alternative with the highest expected payoff.

C7. Uses subjective probabilities and sensitivity analysis to examine how changes in probability assumptions affect the preferred decision.

1 C6 & C7 Expected Monetary Value

When probabilities can be assigned to the possible states of nature, the Expected Monetary Value (EMV) criterion compares alternatives by combining each possible payoff with the probability of the state in which that payoff occurs.

Focus	Expected payoff
First step	Calculate the EMV of each alternative
Second step	Choose the highest EMV

Let $P(S_j)$ represent the probability that state of nature S_j occurs. The probabilities must satisfy

$$\sum_{j=1}^n P(S_j) = 1$$

For each alternative, calculate its Expected Monetary Value:

$$EMV(A_i) = \sum_{j=1}^n P(S_j)P_{ay}(A_i, S_j)$$

For two states of nature, this becomes

$$EMV(A_i) = P(S_1)P_{ay}(A_i, S_1) + P(S_2)P_{ay}(A_i, S_2)$$

Finally, choose the alternative with the highest expected monetary value:

$$A_{\text{Bayes}}^* = \arg \max_i EMV(A_i)$$

Sensitivity Analysis

When the probabilities of the states are subjective, the preferred alternative may depend on the probability assigned to each state. Sensitivity analysis examines how the decision changes when these probabilities change. For two states of nature, let $p = P(S_1)$. Since the probabilities must sum to one,

$$P(S_1) = p, \quad P(S_2) = 1 - p, \quad 0 \leq p \leq 1.$$

The EMV of each alternative can then be written as a function of p :

$$EMV(A_i) = pP_{ay}(A_i, S_1) + (1 - p)P_{ay}(A_i, S_2)$$

A threshold probability occurs when two alternatives have equal expected monetary value:

$$EMV(A_i) = EMV(A_k)$$

The threshold divides the feasible probability range into regions in which different alternatives have the highest EMV. Therefore, the sensitivity analysis determines which alternative is preferred for each range of values of p and identifies any probability at which the decision-maker is indifferent between alternatives.

Section	Type	Criterion
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2 Solved Example: Clothing Store

A local clothing store is deciding whether to launch a new advertising campaign before the holiday season or continue with its current marketing strategy. The manager estimates the probabilities of strong and weak sales based on previous holiday seasons. S_1 : Strong Sales, S_2 : Weak Sales

Alternative	S_1	S_2
$P(S_j)$:	60%	40%
Launch new campaign (A_1)	55	5
Keep current strategy (A_2)	35	22

All payoffs are expressed in thousands of dollars. What decision should the manager make using the Expected Monetary Value criterion?

2.1 Solution

Step 0: Identify the probabilities

The probabilities of the states of nature are $P(S_1) = 0.6$ $P(S_2) = 0.4$

Check they sum to 1: $P(S_1) + P(S_2) = 0.6 + 0.4 = 1$

Step 1.1: Calculate the EMV of A_1

For alternative A_1 , multiply each payoff by the probability of its corresponding state and add the results:

$$\begin{aligned} EMV(A_1) &= P(S_1)P_{ay}(A_1, S_1) + P(S_2)P_{ay}(A_1, S_2) \\ &= (0.6)(55) + (0.4)(5) \\ &= 33 + 2 = 35 \end{aligned}$$

Thus, the expected monetary value of launching the new campaign is \$35,000.

Step 1.2: Calculate the EMV of A_2

For alternative A_2 :

$$\begin{aligned} EMV(A_2) &= P(S_1)P_{ay}(A_2, S_1) + P(S_2)P_{ay}(A_2, S_2) \\ &= (0.6)(35) + (0.4)(22) \\ &= 21 + 8.8 = 29.8 \end{aligned}$$

Thus, the expected monetary value of keeping the current strategy is \$29,800.

Step 2: Apply the Bayes decision criterion

Compare the two expected monetary values:

$$R^* = \max\{35, 29.8\} = 35$$

$$A_{\text{Bayes}}^* = \arg \max_i EMV(A_i) = A_1$$

The Bayes decision criterion selects the new advertising campaign because it has the higher Expected Monetary Value: \$35,000 compared with \$29,800.

Extended payoff table

Alternative	S_1 (0.6)	S_2 (0.4)	EMV
A_1	55	5	$(0.6)(55) + (0.4)(5) = 35$
A_2	35	22	$(0.6)(35) + (0.4)(22) = 29.8$
Highest EMV	35		

$$A_{\text{Bayes}}^* = \arg \max_i EMV(A_i) = A_1$$

3 Practice: Sports Company

A sports equipment company must choose a production strategy for a new product. Demand could be high, medium, or low. Based on market research, the company estimates that there is a 50% probability of high demand, a 30% probability of medium demand, and a 20% probability of low demand.

Alternative	Demand		
	High (S_1)	Medium (S_2)	Low (S_3)
Large production	90	30	-15
Medium production	65	45	20

All payoffs are expressed in thousands of dollars. Determine the decision using the Expected Monetary Value criterion.

Step 1: Calculate the EMV of each alternative

$$\begin{aligned} EMV(A_1) &= (0.5)(90) + (0.3)(30) + (0.2)(-15) \\ &= \underline{\hspace{2cm}} \end{aligned}$$

$$\begin{aligned} EMV(A_2) &= (0.5)(65) + (0.3)(45) + (0.2)(20) \\ &= \underline{\hspace{2cm}} \end{aligned}$$

Step 2: Apply the Bayes decision criterion

Choose the alternative with the highest expected monetary value:

$$EMV^* = \max\{_, _ \} = _$$

$$A_{\text{Bayes}}^* = \arg \max_i EMV(A_i) = _$$

4 Practice: Electronics Company

An electronics company must choose a production strategy for a new product. Demand could be high or low. Based on market research, the company estimates that there is a 60% probability of high demand and a 40% probability of low demand. All payoffs are expressed in thousands of dollars.

Alternative	Demand	
	High (S_1)	Low (S_2)
Large production	88	-12
Medium production	64	14
Small production	42	30
Maintain production	20	20

Step 1: Calculate the EMV of each alternative

Alt.	EMV
A_1	$(0.6)(88) + (0.4)(-12) = _$
A_2	$(0.6)(64) + (0.4)(14) = _$
A_3	$(0.6)(42) + (0.4)(30) = _$
A_4	$(0.6)(20) + (0.4)(20) = _$

Step 2: Apply the Bayes decision criterion

$$EMV^* = \max\{_, _, _, _ \} = _$$

$$A_{\text{Bayes}}^* = \arg \max_i EMV(A_i) = _$$

5 Practice: Coffee Shop

A coffee shop must decide how many pastries to prepare each morning. The possible alternatives are to prepare a large quantity (A_1), a medium quantity (A_2), a small quantity (A_3), or maintain the usual quantity (A_4) of pastries. Demand could be high (S_1), medium (S_2), or low (S_3). Based on historical sales data, the coffee shop estimates that there is a 50% probability of high demand, a 30% probability of medium demand, and a 20% probability of low demand.

Alternative	Demand		
	S_1	S_2	S_3
A_1	72	38	8
A_2	58	45	20
A_3	40	35	28
A_4	22	22	22

What alternative should the coffee shop choose using the Expected Monetary Value criterion?

6 Practice: Technology Company

A technology company must decide how many units of a new product to produce before the start of the next sales period. The possible alternatives are to produce a large quantity (A_1), a medium quantity (A_2), or a small quantity (A_3).

Demand could be very high (S_1), high (S_2), medium (S_3), or low (S_4). Based on market research, the company estimates that there is a 30% probability of very high demand, a 35% probability of high demand, a 20% probability of medium demand, and a 15% probability of low demand.

Alternative	Demand			
	S_1	S_2	S_3	S_4
A_1	95	70	30	-10
A_2	72	58	38	18
A_3	48	45	35	25

What alternative should the company choose using the Expected Monetary Value criterion?

7 Example: Clothing Store

A local clothing store must choose between launching a new advertising campaign (A_1) or keeping its current marketing strategy (A_2) for the holiday season. Sales may be strong (S_1) or weak (S_2). Let p be the subjective probability of strong sales, so weak sales have probability $1 - p$. Payoffs are in thousands of dollars. Using the Expected Value criterion, determine how the preferred alternative changes with p and identify the threshold probability at which the manager is indifferent.

Alternative	S_1	S_2
$P(S_j)$:	p	$1 - p$
Launch new campaign (A_1)	55	5
Keep current strategy (A_2)	45	25

Step 1.1: Calculate the expected value of A_1

For the Launch new campaign alternative:

$$\begin{aligned} EV(A_1) &= p(55) + (1 - p)(5) \\ &= 55p + 5 - 5p = 50p + 5 \end{aligned}$$

Thus, the expected payoff from launching the new campaign is $50p + 5$ thousand dollars.

Step 1.2: Calculate the expected value of A_2

For the Keep current strategy alternative:

$$\begin{aligned} EV(A_2) &= p(45) + (1 - p)(25) \\ &= 45p + 25 - 25p = 20p + 25 \end{aligned}$$

Thus, the expected payoff from keeping the current strategy is $20p + 25$ thousand dollars.

Step 2: Find the threshold probability

The threshold occurs when the two alternatives have the same expected value:

$$EV(A_1) = EV(A_2)$$

$$50p + 5 = 20p + 25 \quad \longrightarrow \quad 30p = 20 \quad \longrightarrow \quad p = \frac{2}{3}$$

Step 3: Perform the sensitivity analysis

Compare the expected values on each side of the threshold.

$$p < \frac{2}{3} \quad \Rightarrow \quad EV(A_1) < EV(A_2) \quad \Rightarrow \quad \text{choose } A_2$$

$$p > \frac{2}{3} \quad \Rightarrow \quad EV(A_1) > EV(A_2) \quad \Rightarrow \quad \text{choose } A_1$$

$$p = \frac{2}{3} \quad \Rightarrow \quad EV(A_1) = EV(A_2) \quad \Rightarrow \quad \text{Indifferent}$$

$$A_{\text{Bayes}}^* = \begin{cases} A_2 & 0 \leq p < \frac{2}{3} \\ \text{Indifferent} & p = \frac{2}{3} \\ A_1 & \frac{2}{3} < p \leq 1 \end{cases}$$

8 Practice: Clothing Store

A local clothing store must decide how to respond to expected sales during the holiday season. The possible alternatives are to launch a new advertising campaign (A_1) or keep the current marketing strategy (A_2). Sales could be strong (S_1) or weak (S_2). Let p represent the subjective probability of strong sales. All payoffs are expressed in thousands of dollars. Using the Expected Value criterion, determine how the preferred alternative changes with p and identify the threshold probability at which the manager is indifferent.

Alternative	S_1	S_2
$P(S_j)$:	p	$1 - p$
Launch new campaign (A_1)	50	10
Keep current strategy (A_2)	40	20

Step 1.1: Calculate the expected value of A_1

For the Launch new campaign alternative:

$$\begin{aligned} EV(A_1) &= p(50) + (1 - p)(10) \\ &= \underline{\hspace{2cm}} \\ &= \underline{\hspace{2cm}} \end{aligned}$$

Thus, the expected payoff from launching the new campaign is $\underline{\hspace{2cm}}$ thousand dollars.

Step 1.2: Calculate the expected value of A_2

For the Keep current strategy alternative:

$$\begin{aligned} EV(A_2) &= p(40) + (1 - p)(20) \\ &= \underline{\hspace{2cm}} \\ &= \underline{\hspace{2cm}} \end{aligned}$$

Thus, the expected payoff from keeping the current strategy is $\underline{\hspace{2cm}}$ thousand dollars.

Step 2: Find the threshold probability

The threshold occurs when the two alternatives have the same expected value:

$$\begin{aligned} EV(A_1) &= EV(A_2) \\ \underline{\hspace{2cm}} &= \underline{\hspace{2cm}} \\ \underline{\hspace{1cm}} p &= \underline{\hspace{1cm}} \quad \rightarrow \quad p = \underline{\hspace{2cm}} \end{aligned}$$

Therefore, the manager is indifferent between the two alternatives when the subjective probability of strong sales is $\underline{\hspace{2cm}}$.

Step 3: Perform the sensitivity analysis

Compare the expected values on each side of the threshold.

$$p < \underline{\hspace{1cm}} \Rightarrow EV(A_1) < EV(A_2) \Rightarrow \text{choose } \underline{\hspace{1cm}},$$

$$p > \underline{\hspace{1cm}} \Rightarrow EV(A_1) > EV(A_2) \Rightarrow \text{choose } \underline{\hspace{1cm}},$$

$$p = \underline{\hspace{1cm}} \Rightarrow EV(A_1) = EV(A_2) \Rightarrow \underline{\hspace{2cm}}.$$

$$\rightarrow A_{\text{Bayes}}^* = \begin{cases} \underline{\hspace{2cm}}, & 0 \leq p < \underline{\hspace{1cm}}, \\ \underline{\hspace{2cm}}, & p = \underline{\hspace{1cm}}, \\ \underline{\hspace{2cm}}, & \underline{\hspace{1cm}} < p \leq 1. \end{cases}$$

9 Practice: Technology Company

A technology company must decide how many units of a new product to produce before the next sales period. The possible alternatives are to produce a large quantity (A_1) or a small quantity (A_2). Demand could be high (S_1) or low (S_2). Let p represent the subjective probability of high demand. All payoffs are expressed in thousands of dollars.

Using the Expected Monetary Value criterion, determine how the preferred alternative changes with p and identify the threshold probability at which the company is indifferent.

Alternative	S_1	S_2
$P(S_j)$:	p	$1 - p$
Produce large quantity (A_1)	70	10
Produce small quantity (A_2)	45	30

10 Practice: Transportation Company

A transportation company must decide how to prepare its fleet for the next operating period. The possible alternatives are to prepare a large fleet (A_1) or a small fleet (A_2). Demand could be high (S_1) or low (S_2). Let p represent the subjective probability of high demand. All payoffs are expressed in thousands of dollars.

Using the Expected Monetary Value criterion, determine how the preferred alternative changes with p and identify the threshold probability at which the company is indifferent.

Alternative	S_1	S_2
$P(S_j)$:	p	$1 - p$
Prepare large fleet (A_1)	50	40
Prepare small fleet (A_2)	45	35

11 Practice: Restaurant Company

A restaurant company must decide how many meals to prepare before the next operating period. The possible alternatives are to prepare a large quantity (A_1), a medium quantity (A_2), or a small quantity (A_3). Demand could be high (S_1) or low (S_2). Let p represent the subjective probability of high demand. All payoffs are expressed in thousands of dollars.

Using the Expected Monetary Value criterion, determine how the preferred alternative changes with p and identify the threshold probabilities at which the company is indifferent between alternatives.

Alternative	S_1	S_2
$P(S_j)$:	p	$1 - p$
Large quantity (A_1)	80	10
Medium quantity (A_2)	60	30
Small quantity (A_3)	40	35

12 Practice: Manufacturing Company

A manufacturing company must decide how many units of a new product to produce before the next sales period. The possible alternatives are to produce a large quantity (A_1) or a small quantity (A_2).

Demand could be high (S_1), medium (S_2), or low (S_3). Let p represent the subjective probability of high demand. The probabilities of medium and low demand are $0.4(1 - p)$ and $0.6(1 - p)$, respectively. All payoffs are expressed in thousands of dollars.

Using the Expected Monetary Value criterion, determine how the preferred alternative changes with p and identify the threshold probability at which the company is indifferent.

Alternative	S_1	S_2	S_3
$P(S_j)$:	p	$0.4(1 - p)$	$0.6(1 - p)$
Large quantity (A_1)	80	55	5
Small quantity (A_2)	50	40	25

13 Practice: Electronics Retailer

An electronics retailer must decide how many units of a new product to order before the next sales period. The possible alternatives are to order a very large quantity (A_1), a large quantity (A_2), a medium quantity (A_3), or a small quantity (A_4). Demand could be high (S_1) or low (S_2). Let p represent the subjective probability of high demand. All payoffs are expressed in thousands of dollars.

Using the Expected Monetary Value criterion, determine how the preferred alternative changes with p and identify the threshold probabilities at which the company is indifferent between alternatives.

Alternative	S_1	S_2
$P(S_j)$:	p	$1 - p$
Very large quantity (A_1)	90	5
Large quantity (A_2)	75	25
Medium quantity (A_3)	55	40
Small quantity (A_4)	35	45

14 Practice: Concert Organizer

A concert organizer must decide how many tickets to make available for an upcoming event. The possible alternatives are to release a large number of tickets (A_1) or a small number of tickets (A_2).

Attendance could be high (S_1), medium (S_2), or low (S_3). Let p represent the subjective probability of high attendance and q the subjective probability of medium attendance. The probability of low attendance is $1 - p - q$. All payoffs are expressed in thousands of dollars.

Using the Expected Monetary Value criterion, determine how the preferred alternative changes with p and q and identify the combinations of p and q at which the organizer is indifferent between alternatives.

Alternative	S_1	S_2	S_3
$P(S_j)$:	p	q	$1 - p - q$
Large number of tickets (A_1)	80	55	5
Small number of tickets (A_2)	50	40	25